



2015

TRIAL HSC EXAMINATION

# Mathematics

## General Instructions

- Reading Time - 5 minutes
- Working Time - 3 hours
- Write using a blue or black pen. Black pen is preferred
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-16

Total marks

95

### **Section I**

Total marks (10)

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided on the reverse side of this page.
- Allow about 15 minutes for this section

### **Section II**

Total marks

85

- Attempt questions 11 – 16
- Answer on the blank paper provided, unless otherwise instructed
- Start a new page for each question
- All necessary working should be shown for every question
- Allow about 2 hour 45 minutes for this section

**Section I****10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section**

Use the multiple choice answer sheet provided for Questions 1 – 10.

1. Evaluate  $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$ .

- (A) 0
- (B) undefined
- (C) 4
- (D) 1

2. Simplify :  $\frac{\cos\left(\frac{\pi}{2} - \theta\right)}{\sin\left(\frac{\pi}{2} - \theta\right)}$

- (A) 1
- (B)  $\cot\theta$
- (C)  $\tan\theta$
- (D)  $-\tan\theta$

3. The quadratic equation  $x^2 - kx + 3 = 0$  has roots  $\alpha$  and  $\beta$ .

What is the value of  $\alpha^2\beta + \beta^2\alpha$ ?

- (A)  $k - 3$
- (B)  $k + 3$
- (C)  $-3k$
- (D)  $3k$

4. Which of the following is a primitive of the function  $5 - \frac{1}{e^x}$ ?

- (A)  $5x + \frac{1}{e^x} + C$
- (B)  $5x - \frac{1}{e^x} + C$
- (C)  $5 - \frac{1}{e^{2x}} + C$
- (D)  $5x - \ln x + C$

5. Differentiate  $-\frac{1}{2x^2}$

- (A)  $\frac{4}{x^3}$   
(B)  $\frac{1}{x^3}$   
(C)  $\frac{1}{x}$   
(D)  $-\frac{1}{x^3}$

6. The line  $y = mx + b$  is a tangent to the curve  $y = x^3 - 3x + 2$  at the point  $(-2, 0)$ . What are the values of  $m$  and  $b$ ?

- (A)  $m = 9$  and  $b = -18$   
(B)  $m = 9$  and  $b = 18$   
(C)  $m = 12$  and  $b = -18$   
(D)  $m = 12$  and  $b = 18$

7. The table below shows the values of a function  $f(x)$  for five values of  $x$ .

|        |   |     |    |     |   |
|--------|---|-----|----|-----|---|
| $x$    | 2 | 2.5 | 3  | 3.5 | 4 |
| $f(x)$ | 4 | 1   | -2 | 3   | 8 |

What value is an estimate for  $\int_2^4 f(x)dx$  using Simpson's rule with these five values?

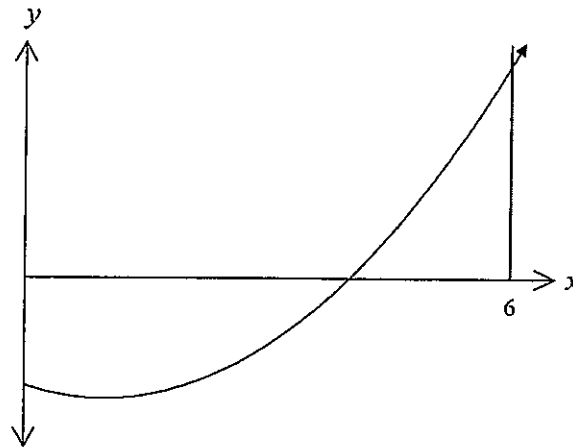
- (A) 4  
(B) 6  
(C) 8  
(D) 12

8. Ten kilograms of chlorine are placed in water and begin to dissolve. After  $t$  hours the amount  $A$  kg of undissolved chlorine is given by  $A = 10e^{-kt}$ . What is the value of  $k$  given that  $A = 3.6$  and  $t = 5$ ?

- (A)  $-0.717$   
(B)  $-0.204$   
(C)  $0.204$

(D) 0.717

9. The diagram below shows the graph of  $y = x^2 - 2x - 8$ .



What is the correct expression for the area bounded by the  $x$ -axis and the curve  $y = x^2 - 2x - 8$  between  $0 \leq x \leq 6$ ?

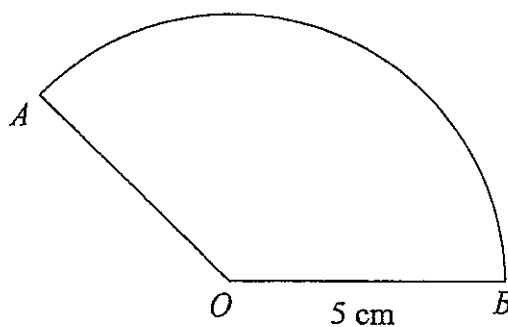
(A)  $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$

(B)  $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$

(C)  $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$

(D)  $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$

10.  $AOB$  is a sector of a circle, centre  $O$  and radius 5 cm. The sector has an area of  $10\pi$ .



Not to scale

What is the arc length of the sector?

- (A)  $2\pi$
- (B)  $4\pi$
- (C)  $6\pi$
- (D)  $10\pi$

**End of Section I****Section II****Total marks (60)****Attempt Questions 11-16****Allow about 1 hour 45 minutes for this section**

Answer all questions, starting each question on a new sheet of paper with your name and question number at the top of the page. Do not write on the back of sheets.

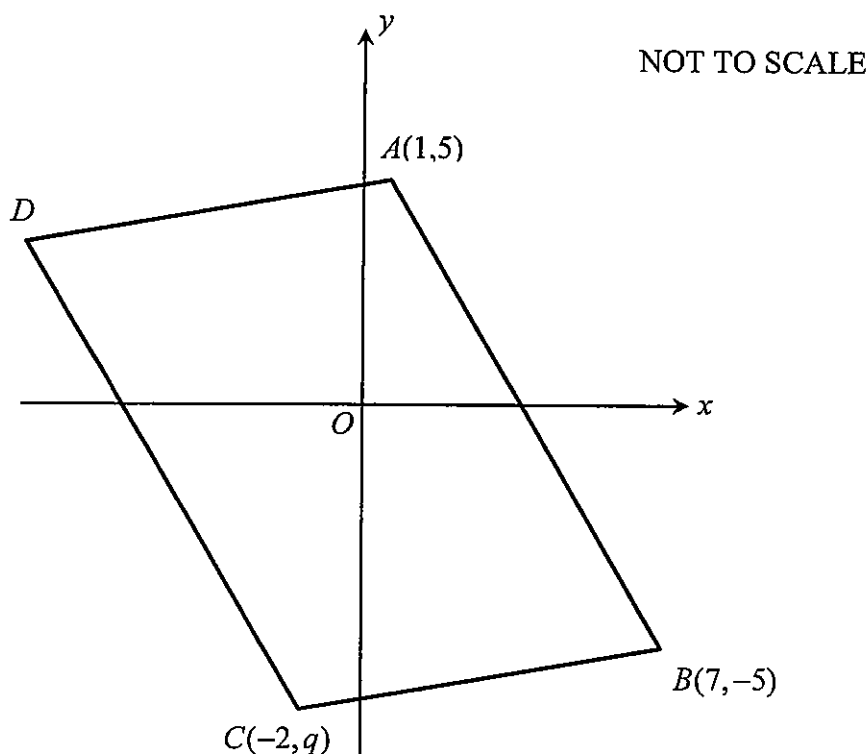
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| <b>Question 11 (10 marks)</b>                                                                                                         | <b>Use a separate sheet of paper</b> | <b>Marks</b> |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------|
| a) Factorise: $m^2 - n^2 + 5m - 5n$                                                                                                   |                                      | 2            |
| b) Find in radians correct to three significant figures, the acute angle between the line $2x - 3y - 5 = 0$ and the positive $x$ axis |                                      | 2            |
| c) Simplify $\frac{4^{3n} \times 16^{1-3n}}{8^{-2n}}$                                                                                 |                                      | 2            |
| d) Differentiate $\frac{5}{\sqrt{2-3x^2}}$                                                                                            |                                      | 2            |
| e) Express $6\sqrt{5} - \frac{1}{\sqrt{5}-2}$ in the form $a + b\sqrt{5}$ where $a$ and $b$ are integers.                             |                                      | 2            |

**End Question 11**

**Question 12 (15 marks)** Use a separate sheet of paper

- a) In the diagram below  $A$ ,  $B$ , and  $C$  have coordinates  $(1, 5)$ ,  $(7, -5)$  and  $(-2, q)$  respectively.  $C$  is in the third quadrant and  $ABCD$  is a parallelogram.

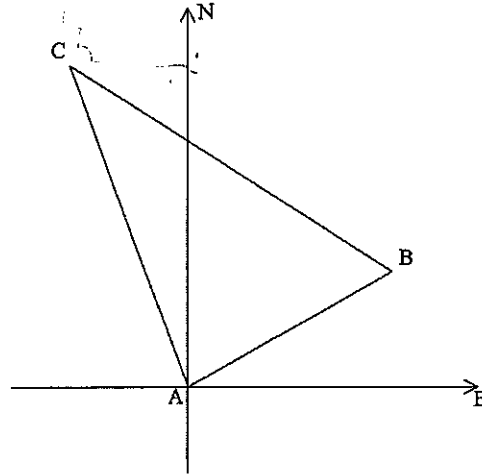


- |       |                                                                                                         |   |
|-------|---------------------------------------------------------------------------------------------------------|---|
| (i)   | Show that the equation of $AB$ is $5x + 3y - 20 = 0$ .                                                  | 2 |
| (ii)  | Write down an expression, in terms of $q$ , for the perpendicular distance from $C$ to the line $AB$ .  | 1 |
| (iii) | Find the length of the interval $AB$ .                                                                  | 1 |
| (iv)  | Given that the area of $ABCD$ is 100 square units, find the value of $q$ .                              | 3 |
|       |                                                                                                         |   |
| b)    | If the points $(-2a, 3)$ , $(a - 1, a - 2)$ and $(a - 3, a + 1)$ are collinear, find the value of $a$ . | 2 |
|       |                                                                                                         |   |
| c)    | Differentiate $\sin(2x) + \cos\left(\frac{x}{2}\right)$                                                 | 2 |
|       |                                                                                                         |   |
| d)    | Find $\int \frac{x^3 + x}{2x^2} dx$                                                                     | 2 |
|       |                                                                                                         |   |
| e)    | By completing the squares or otherwise find the vertex of the parabola $x^2 - 10x + 15 = 2y$            | 2 |

**End Question 12**

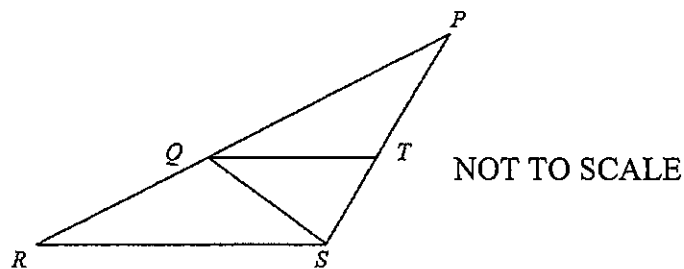
**Question 13 (15 marks)** Use a separate sheet of paper**Marks**

- a) A ship sails 50 km from Port  $A$  to Port  $B$  on a bearing of  $63^\circ$ , then sails 130 km from Port  $B$  to Port  $C$ , on a bearing of  $296^\circ$ .



- |      |                                                                             |   |
|------|-----------------------------------------------------------------------------|---|
| i)   | Copy the diagram and mark on it all the given information.                  | 1 |
| ii)  | Show that $\angle ABC = 53^\circ$ .                                         | 1 |
| iii) | Find, to the nearest km, the distance of Port $A$ from Port $C$ .           | 1 |
| iv)  | Find the bearing of Port $A$ from Port $C$ , correct to the nearest degree. | 2 |

- b) In the diagram,  $QT \parallel RS$  and  $TQ$  bisects  $\angle PQS$ .



Copy the diagram into your answer booklet, showing this information.

- |      |                                          |   |
|------|------------------------------------------|---|
| i)   | Explain why $\angle TQS = \angle QSR$ .  | 1 |
| ii)  | Prove that $\triangle QRS$ is isosceles. | 2 |
| iii) | Hence show that $PT : TS = PQ : QS$      | 2 |

**Question 13 continued****Marks**

c) Solve for  $x$  :  $\log_2(x+1) + \log_2(x+3) = 3$

**3**

d) In a geometric sequence the third term is 18 and the seventh term is 1458.

Find the first term and the common ratio.

**2****End Question 13.**

**Question 14 (15 marks)**      Use a separate sheet of paper **Marks**

a) i) Show that  $\frac{d}{dx} \{ \log_e x \}^2 = \frac{2 \log_e x}{x}$  **1**

ii) Hence evaluate  $\int_1^e \frac{\log_e x}{x} dx$  **1**

b) A rain water tank which is full is drained so that at time 't' minutes, the volume of water  $V$  in litres is given by

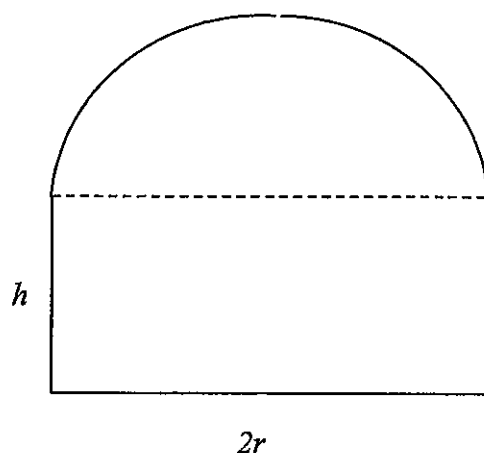
$$V = 500 \left( 1 - \frac{t}{60} \right)^2 \quad \text{for } 0 \leq t \leq 60.$$

i) How much water was initially in the tank? **1**

ii) After how many minutes was the tank half full? **2**

iii) At what rate was the water draining when the time is 58 min. **2**

c) Lisa has designed a garden bed which consists of a rectangle and a semicircle as shown in the diagram. If the perimeter of the garden bed is to be 20 metres:



i. Find an expression for  $h$  in terms of  $r$ . **1**

ii. Show that the area of the garden bed can be given by the formula **1**

$$A = 20r - 2r^2 - \frac{1}{2}\pi r^2.$$

iii. Find the value of  $r$  that gives the maximum area and find this area. **3**

d) Consider the function  $f(x) = \frac{x^2}{1+x^2}$ . **3**

Show that the second derivative of this function is  $f''(x) = \frac{2(1-3x^2)}{(1+x^2)^3}$

and hence find the values of  $x$  for which this function is concave up.

**Question 15 (15 marks)** Use a separate sheet of paper**Marks**

- a) A quantity of radioactive material decays at a rate proportional to the amount,  $M$ , present at any time,  $t$ .

- (i) Given that  $M = M_0 e^{-kt}$  represents the mass of material in grams at any time  $t$  years after the material was produced, show that

**1**

$$\frac{dM}{dt} = -kM$$

- (ii) If initially there was 3500 g of material and after 4 years the mass had decayed to 2300 g, calculate  $k$ , correct to 4 significant figures.

**2**

- (iii) Determine the number of years needed for the material to decay to 25% of its original quantity.

**2**

- b) A particle moves in a straight line so that its velocity,  $v$  metres per second, at time  $t$  is given by  $v = 3 - \frac{2}{1+t}$ .

The particle is initially 1 metre to the right of the origin.

- (i) Find an expression for the position  $x$ , of the particle at time  $t$ .
- (ii) Explain why the velocity of the particle is never 3 metres per second.
- (iii) Find the acceleration of the particle when  $t = 2$  seconds.

**2****1****2**

- c) Consider the quadratic equation in  $x$ :

$$(p^2 + q^2)x^2 + 2q(p+r)x + (q^2 + r^2) = 0$$

**3**

Find an equation in terms of  $p$ ,  $q$  and  $r$  such that the quadratic has real roots

- d) Give a possible sketch for a curve which has the following defining features:

**2**

When  $x = -5, 0$  and  $3$ ,

$$\frac{dy}{dx} = 0.$$

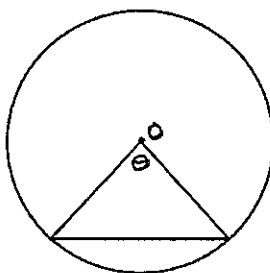
Also  $\frac{d^2y}{dx^2} > 0$  for  $x < -2\frac{1}{2}$  and  $x > 1\frac{1}{2}$ , while  $\frac{d^2y}{dx^2} < 0$  for  $-2\frac{1}{2} < x < 1\frac{1}{2}$ .

**Question 16 (15 marks)** Use a separate sheet of paper

- a) *TempIT* is an employment agency which specialises in contracting temporary employees. They have analysed the number of job applications received over the last five years. They found that the demand ( $D$ ), measured in hundreds, for temporary employment at time ( $t$  years) is given by the function:

$$D(t) = 4 \sin\left(\frac{\pi}{4}t\right) + 7$$

- i) Find all the times in the next 12 years where demand will be at its peak 3
- ii) State the amplitude and period of  $D(t)$  and sketch its graph for the first twelve years. 3
- b) The area of a circle radius  $r$  cm, is divided in the ratio 15:7 by a chord which subtends an angle of  $\theta$  radians at the centre of the circle.



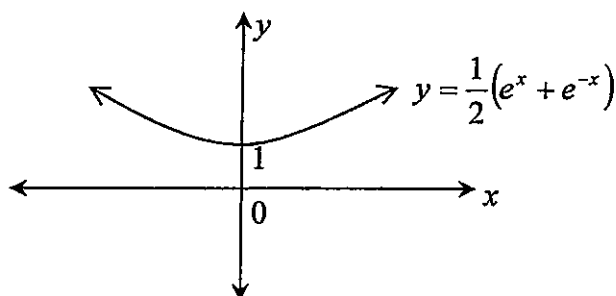
- i) Show that the ratio of the areas of the major segment to minor segment is given by:

$$\frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta} \quad 2$$

- ii) Given that  $\pi = \frac{22}{7}$  prove that  $\theta = 2 + \sin \theta$  1

**Question 16 continues next page**

- c) The sketch of the catenary curve  $y = \frac{1}{2}(e^x + e^{-x})$  is given below. A catenary curve is the shape obtained when a chain or rope is strung between two points.



Calculate the volume of the solid generated when the curve

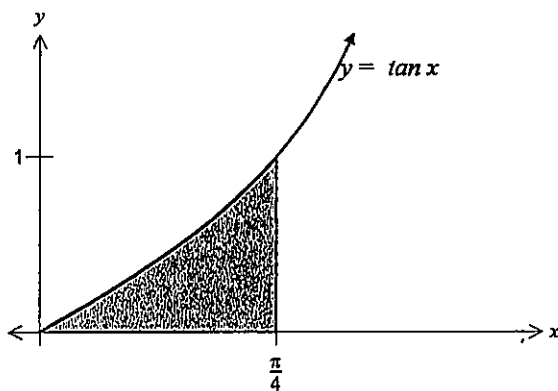
3

$$y = \frac{1}{2}(e^x + e^{-x})$$

is rotated about the x-axis between the ordinates  $x = -3$  and  $x = 3$ .

- d) The area bounded by the curve  $y = \tan x$ , the lines  $x = 0$  and  $x = \frac{\pi}{4}$  and the x-axis is shaded below.

3



Evaluate this area correct to 3 significant figures

**End of Examination**

## Trial HSC– Mathematics - 2015

## Section I – Multiple Choice Answer Sheet

Name \_\_\_\_\_

**Allow about 15 minutes for this section**

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.A ☒ B ☒ <sup>correct</sup> C ☐ D ☐Start  
Here

- 1. A ☐ B ☐ C ☒ D ☐  
2. A ☐ B ☐ C ☒ D ☐  
3. A ☐ B ☐ C ☐ D ☒  
4. A ☒ B ☐ C ☐ D ☐  
5. A ☐ B ☒ C ☐ D ☐  
6. A ☐ B ☒ C ☐ D ☐  
7. A ☒ B ☐ C ☐ D ☐  
8. A ☐ B ☐ C ☒ D ☐  
9. A ☐ B ☐ C ☐ D ☒  
10. A ☐ B ☒ C ☐ D ☐

$$1) \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$$

$$\lim_{x \rightarrow 3} x+1 = 4 \quad (C)$$

$$2) \frac{\cos(\frac{\pi}{2} - \theta)}{\sin(\frac{\pi}{2} - \theta)} = \frac{\sin \theta}{\cos \theta} = \tan \theta \quad (C)$$

$$3) \alpha \beta (\alpha + \beta) \\ \frac{c}{a} \times \frac{-b}{a} \\ 3 \times k = 3k \quad (D)$$

$$4) \int 5 - e^{-x} dx \\ = 5x + e^{-x} + C \quad (A)$$

$$5) \frac{d}{dx} -\frac{1}{2}x^{-2} = x^{-3} = \frac{1}{x^3} \quad (B)$$

$$6) \frac{dy}{dx} = 3x^2 + 3 \\ \text{at } x = -2 \quad m = 12 + 3 = 9$$

$$y - 0 = 9(x + 2)$$

$$y = 9x + 18$$

$$m = 9 \quad b = 18 \quad (B)$$

$$7) \frac{15}{3} (4 + 8 + 4(1 + 3) + 2 \times -2)$$

$$\frac{15}{3} (12 + 16 - 4)$$

$$4 \quad (A)$$

$$8) A = 10e^{-kt}$$

$$3.6 = 10e^{-5k}$$

$$\ln(0.36) = -5k$$

$$k = 0.204 \quad (C)$$

$$9) (x-4)(x+2) = 0$$

cuts at  $x = 4$  and  $-2$

$$\left| \int_0^4 \right| + \left| \int_4^6 \right| \quad (D)$$

$$10) \frac{1}{2} r^2 \theta = 10\pi$$

$$\frac{25}{2} \theta = 10\pi$$

$$\theta = \frac{4}{5} \pi$$

$$L = r \theta$$

$$= 5 \times \frac{4}{5} \pi$$

$$= 4\pi \quad (B)$$



# QUESTION 11

4.

$$a) (m-n)(m+n) + 5(m-n) \\ = (m-n)(m+n+5)$$

$$b) \tan \theta = m$$

$$2x - 5 = 3y$$

$$\frac{2}{3}x - \frac{5}{3} = y$$

$$m = \frac{2}{3}$$

$$\theta = 0.588.$$

$$c) \frac{2^{6n} \times 2^{4(1-3n)}}{(2^3)^{-2n}}$$

$$= \frac{2^{6n} \times 2^{4-12n}}{2^{-6n}}$$

$$= \frac{2^{4-6n}}{2^{-6n}}$$

$$= 2^4$$

$$d) 5(2-3x^2)^{-1/2}$$

$$\frac{dy}{dx} = \frac{-1}{2} \times 5(2-3x^2)^{-3/2} \times -6x \quad \text{Accept to here}$$

$$= \frac{15x}{\sqrt{(2-3x^2)^3}}$$

$$e) 6\sqrt{5} - \frac{1}{\sqrt{5-2}} = 6\sqrt{5} - \frac{\sqrt{5+2}}{5-4}$$

$$= 6\sqrt{5} - \sqrt{5} - 2$$

$$= -2 + 5\sqrt{5}$$

$$a = -2$$

$$b = 5$$

# QUESTION 12.

3.

$$\begin{aligned} \text{a) i) } m &= \frac{5 - -5}{1 - 7} \\ &= \frac{10}{-6} \\ &= -\frac{5}{3} \end{aligned}$$

$$y - 5 = -\frac{5}{3}(x - 1)$$

$$3y - 15 = -5x + 5$$

$$5x + 3y - 20 = 0$$

$$\begin{aligned} \text{ii) } d &= \frac{|5(-2) + 3q - 20|}{\sqrt{5^2 + 3^2}} \\ &= \frac{|3q - 30|}{\sqrt{34}} \end{aligned}$$

$$\begin{aligned} \text{iii) } AB &= \sqrt{6^2 + 10^2} \\ &= \sqrt{136} \\ &= 2\sqrt{34} \end{aligned}$$

$$\text{iv) Area} = b \times h$$

$$\frac{2\sqrt{34} \times |3q - 30|}{\sqrt{34}} = 100$$

$$|3q - 30| = 50$$

$$\therefore 3q - 30 = 50 \quad \text{or} \quad 3q - 30 = -50$$

$$3q = 80 \quad \text{or} \quad 3q = -20$$

$$q = 26\frac{2}{3}$$

$$q = -\frac{20}{3}$$

since  $q$  in 3rd quad  $q = -\frac{20}{3}$

$$b) \frac{y_2 - y_1}{x_2 - x_1} = \frac{a-2-3}{a-1+2a} = \frac{a+1-3}{a-3+2a}$$

$$\therefore \frac{a-5}{3a-1} = \frac{a-2}{3a-3}$$

$$(3a-3)(a-5) = (a-2)(3a-1)$$

$$3a^2 - 18a + 15 = 3a^2 - 7a + 2$$

$$11a = 13$$

$$a = \frac{13}{11}$$

$$c) 2 \cos 2x - \frac{1}{2} \sin\left(\frac{x}{2}\right)$$

$$d) \int \frac{x^3}{2x^2} + \frac{x^1}{2x^2} dx$$

$$= \int \frac{x}{2} + \frac{1}{2x} dx$$

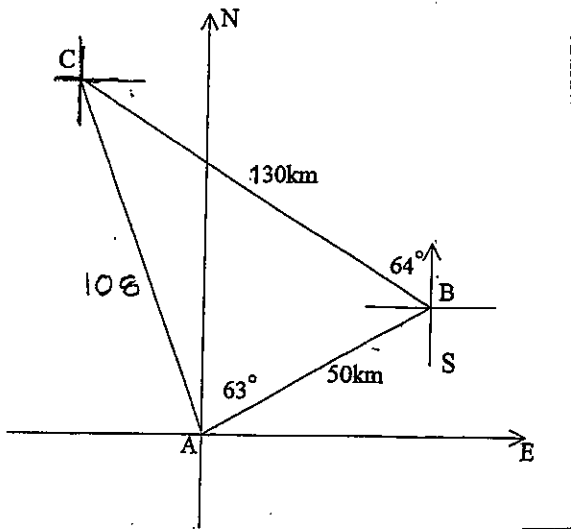
$$= \frac{x^2}{4} + \frac{1}{2} \ln x + C$$

$$e) x^2 - 10x = 2y - 15$$

$$x^2 - 10x + 25 = 2y - 15 + 25$$

$$(x-5)^2 = 2(y+5)$$

$$\text{vertex } (5, -5)$$



ii)  $\angle ABS = 63^\circ$  (alt  $\angle$ 's)

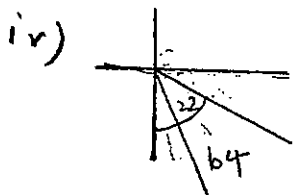
$$\angle ABC = 296 - (180 + 63)$$

$$= 53^\circ$$

iii)  $AC^2 = 130^2 + 50^2 - 2 \times 130 \times 50 \times \cos 53$

$$= 11567$$

$$AC = 108 \text{ km}$$



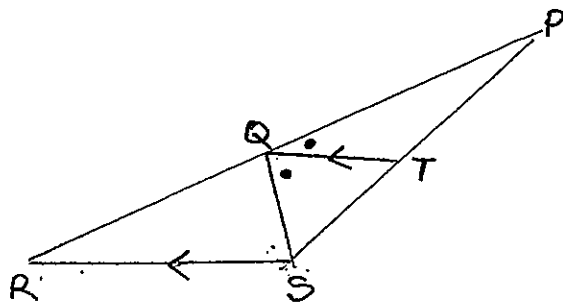
$$\frac{\sin C}{50} = \frac{\sin 53}{108}$$

$$C = 22^\circ$$

$\therefore$  bearing is  $180 + (64 - 22)$

$$= 138^\circ$$

b) .



i)  $\angle TQS = \angle QSR$   
equal alternate angles  $QT \parallel RS$

ii)

$$ii) \angle QRS = \angle PQT \text{ (corresponding } \angle \text{ equal } QT \parallel RS) \quad 6,$$

$$\text{and } \angle PQT = \angle TQS = \angle QSR$$

$$\therefore \angle QRS = \angle QSR$$

$\therefore \triangle QRS$  isosceles (equal base  $\angle$ 's)

$$iii) \text{ Draw line through P } \parallel \text{ to QT.}$$

$$PT : TS = PQ : QR \text{ (ratio of intercepts on } \parallel \text{ lines)}$$

$$\text{but } QR = QS$$

$$\therefore PT : TS = PQ : QS$$

$$c) \log_2 (x+1)(x+3) = 3$$

$$\therefore (x+1)(x+3) = 2^3$$

$$x^2 + 4x + 3 = 8$$

$$x^2 + 4x - 5 = 0$$

$$(x+5)(x-1) = 0$$

$$x = 1 \text{ as } x > 0$$

$$d) T_3 = ar^2 \quad T_7 = ar^6$$

$$18 = ar^2$$

$$1458 = ar^6$$

$$\frac{1458}{18} = \frac{ar^6}{ar^2}$$

$$81 = r^4$$

$$r = \pm 3$$

$$ar^2 = 18$$

$$a \times 9 = 18$$

$$a = 2$$

$$a) \frac{d}{dx} \left\{ \log_e x \right\}^2 = 2 \cdot \log_e x \cdot \frac{1}{x}$$

$$= \frac{2 \log_e x}{x}$$

$$ii) \int_1^e \frac{\log_e x}{x} = \left[ \frac{1}{2} (\log_e x)^2 \right]_1^e$$

$$= \frac{1}{2} (\log_e e)^2 - \frac{1}{2} (\log_e 1)^2$$

$$= \frac{1}{2} \times (\log_e e)^2 - \frac{1}{2} \times 0$$

$$= \frac{1}{2}$$

b)  $t = 0$

i)  $V = 500$

ii)  $V = 250$  for half.

$$250 = 500 \left( 1 - \frac{t}{60} \right)^2$$

$$\frac{1}{2} = \left( 1 - \frac{t}{60} \right)^2$$

$$\pm \sqrt{\frac{1}{2}} = 1 - \frac{t}{60}$$

$$\frac{t}{60} = 1 - \frac{1}{\sqrt{2}}$$

$$t = 0.29 \times 60$$

$$= 17.57 \text{ min.}$$

or  $102.42 \text{ min} - \text{reject}$   
 $(60 - 30\sqrt{2})$

iii)  $V = 500 \left( 1 - \frac{t}{60} \right)^2$

$$\frac{dV}{dt} = 1000 \left( 1 - \frac{t}{60} \right) \times -\frac{1}{60}$$

$$= \frac{-1000}{60} \left( 1 - \frac{t}{60} \right)$$

$$t = 58 \quad \frac{dV}{dt} = \frac{-1000}{60} \left( 1 - \frac{58}{60} \right)$$

$$= \frac{5}{9} \text{ Litres/min}$$

$$c) P = 2r + 2h + \frac{1}{2} \times 2\pi r$$

$$20 = 2r + 2h + \pi r$$

$$\frac{20 - 2r - \pi r}{2} = h$$

$$\text{or } 10 - r = \frac{\pi r}{2} = h$$

$$ii) A = L \times b + \frac{1}{2} \pi r^2$$

$$= 2rh + \frac{1}{2} \pi r^2$$

$$= 2r \left( 10 - r - \frac{\pi r}{2} \right) + \frac{1}{2} \pi r^2$$

$$= 20r - 2r^2 - \pi r^2 + \frac{1}{2} \pi r^2$$

$$= 20r - 2r^2 - \frac{1}{2} \pi r^2$$

$$iii) \frac{dA}{dr} = 20 - 4r - \pi r$$

for a max  $\frac{dA}{dr} = 0$

$$4r + \pi r = 20$$

$$r = \frac{20}{4 + \pi}$$

$$\approx 2.8$$

for a max

$$\frac{d^2A}{dr^2} < 0$$

$$\text{here } \frac{d^2A}{dr^2} = -4 - \pi$$

$$\frac{d^2A}{dr^2} = -7.14$$

$$< 0$$

$\therefore$  max

$$\text{max Area} = 20 \times 2.8 - 2(2.8)^2 - \frac{1}{2} \times \pi \times 2.8^2$$

$$\approx 28 \text{ m}^2$$

$$\text{max } A = \frac{400}{4 + \pi} - \frac{4 + \pi}{2} \cdot \frac{400}{(4 + \pi)^2}$$

$$= \frac{400}{4 + \pi} - \frac{200}{4 + \pi}$$

$$= \frac{200}{4 + \pi} u^2$$

$$) f(x) = \frac{x^2}{1 + x^2}$$

$$f'(x) = \frac{(1 + x^2) \cdot 2x - x^2 \cdot (2x)}{(1 + x^2)^2} \checkmark$$

$$= \frac{2x + 2x^3 - 2x^3}{(1 + x^2)^2}$$

$$= \frac{2x}{(1 + x^2)^2}$$

$$f''(x) = \frac{(1 + x^2)^2 \cdot 2 - 2x \cdot 2 \cdot 2x(1 + x^2)}{(1 + x^2)^4}$$

$$= \frac{2(1 + x^2)[(1 + x^2) - 4x^2]}{(1 + x^2)^4}$$

$$= \frac{2(1 - 3x^2)}{(1 + x^2)^3} \checkmark$$

concave up when  $f''(x) > 0$

$$2(1 - 3x^2) > 0$$

$$1 - 3x^2 > 0$$

$$3x^2 < 1$$

$$x^2 < \frac{1}{3}$$

$$-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

$$1) M = M_0 e^{-kt}$$

$$\frac{dM}{dt} = -k M_0 e^{-kt}$$

$$= -k M \text{ (since } M = M_0 e^{-kt} \text{)}$$

$$\text{ii) } t=0 \quad M_0 = 3500$$

$$2300 = 3500 e^{-4k}$$

$$\frac{23}{35} = e^{-4k}$$

$$k = -\frac{1}{4} \ln\left(\frac{23}{35}\right)$$

$$= 0.10496$$

$$= 0.1050 \text{ (4 sf)}$$

$$\text{iii) } 25\% \text{ of } 3500 = 875 \text{ g}$$

$$875 = 3500 e^{-0.1050t}$$

$$-0.1050t = \ln\left(\frac{1}{4}\right)$$

$$t = 13.2$$

will decay during 14<sup>th</sup> year.

$$\text{b) } v = 3 - \frac{2}{1+t} \quad t=0 \quad x=1$$

$$x = 3t - 2 \ln(1+t) + c$$

$$1 = 0 - 2 \ln 1 + c$$

$$\therefore c = 1$$

$$x = 3t - 2 \ln(1+t) + 1$$

$$\text{ii) since } \frac{2}{1+t} > 0$$

$$\text{then } v = 3 - \frac{2}{1+t} \neq 3$$

$$\text{iii) } v = 3 - 2(1+t)^{-1}$$

$$\frac{dv}{dt} = a = 2(1+t)^{-2}$$

$$t=2 \quad a = \frac{2}{9}$$

$$\text{c) real roots } \Delta \geq 0$$

$$b^2 - 4ac \geq 0$$

$$[2q(p+r)]^2 - 4(p^2+q^2)(q^2+r^2) \geq 0$$

$$4q^2(p^2+r^2+2pr) - 4(p^2q^2+p^2r^2+q^4+q^2r^2) \geq 0$$

$$4q^2p^2 + 4q^2r^2 + 8prq^2 - 4p^2q^2 - 4p^2r^2 - 4q^4 - 4q^2r^2 \geq 0$$

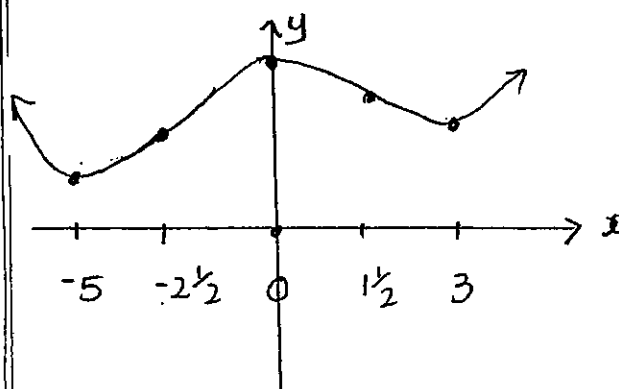
$$8prq^2 - 4p^2r^2 - 4q^4 \geq 0$$

$$q^4 - 2prq^2 + p^2r^2 \leq 0$$

$$(q^2 - pr)^2 \leq 0$$

$$\therefore \text{ as this is a square } q^2 - pr = 0$$

d)



# QUESTION 16

a)  $D(t) = 4 \sin\left(\frac{\pi}{4}t\right) + 7$

find max when  $D'(t) = 0$

$$D'(t) = \pi \cos\left(\frac{\pi}{4}t\right)$$

$$0 = \pi \cos\left(\frac{\pi}{4}t\right)$$

$$\therefore \cos\left(\frac{\pi}{4}t\right) = 0$$

$$\text{when } \frac{\pi}{4}t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$t = 2, 6, 10 \text{ years}$$

for max  $D''(t) \leq 0$

$$D''(t) = -\frac{\pi^2}{4} \sin\left(\frac{\pi}{4}t\right)$$

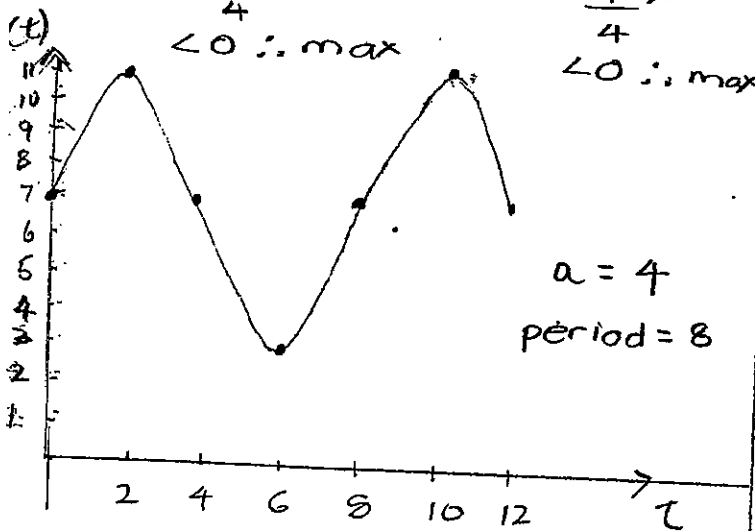
$\therefore$  max at  $t = 2$  and  $t = 10$

$$D''(t) = -\frac{\pi^2}{4} \times 1$$

$< 0 \therefore \text{max}$

$$-\frac{\pi^2}{4} \times 1$$

$< 0 \therefore \text{max}$



b) Area minor =  $\frac{1}{2} r^2 (\theta - \sin \theta)$

Area major =

$$\pi r^2 - \left(\frac{1}{2} r^2 (\theta - \sin \theta)\right)$$

$$\therefore \text{ratio } \frac{\pi r^2 - \frac{1}{2} r^2 \theta + \frac{1}{2} r^2 \sin \theta}{\frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta}$$

10

$$= \frac{2\pi r^2 - \theta^2 \theta + r^2 \sin \theta}{r^2 \theta - r^2 \sin \theta}$$

$\div r^2$

$$= \frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta}$$

ii)  $\frac{15}{7} = \frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta}$

$$15\theta - 15\sin \theta = 14\pi - 7\theta + 7\sin \theta$$

$$22\theta = 14\pi + 22\sin \theta$$

$$\text{if } \pi = \frac{22}{7}$$

$$22\theta = 14 \times \frac{22}{7} + 22\sin \theta$$

$$\theta = 2 + \sin \theta$$

c)  $\pi \int_{-3}^3 y^2 dx$  or  $2\pi \int_0^3 y^2 dx$

$$\therefore 2\pi \int_0^3 \frac{1}{4} (e^{2x} + e^{-2x} + 2) dx$$

$$\frac{\pi}{2} \int_0^3 e^{2x} + e^{-2x} + 2 dx$$

$$\frac{\pi}{2} \left[ \frac{e^{2x}}{2} - \frac{e^{-2x}}{2} + 2x \right]_0^3$$

$$\frac{\pi}{2} \left[ \frac{e^6}{2} - \frac{e^{-6}}{2} + 6 - \left( \frac{e^0}{2} - \frac{e^0}{2} \right) \right]$$

$$\frac{\pi}{2} \left[ \frac{e^6}{2} - \frac{e^{-6}}{2} + 6 \right]$$

$$\underline{\underline{= \frac{\pi}{2} \left[ \frac{e^6 - e^{-6} + 12}{2} \right]}}$$

$$d) \int_0^{\frac{\pi}{4}} \tan x \, dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} \, dx$$

$$= \left[ -\ln(\cos x) \right]_0^{\frac{\pi}{4}}$$

$$= -\ln\left(\cos \frac{\pi}{4}\right) + \ln(\cos 0)$$

$$= -\ln \frac{1}{\sqrt{2}} + \ln 1$$

$$= 0.347$$